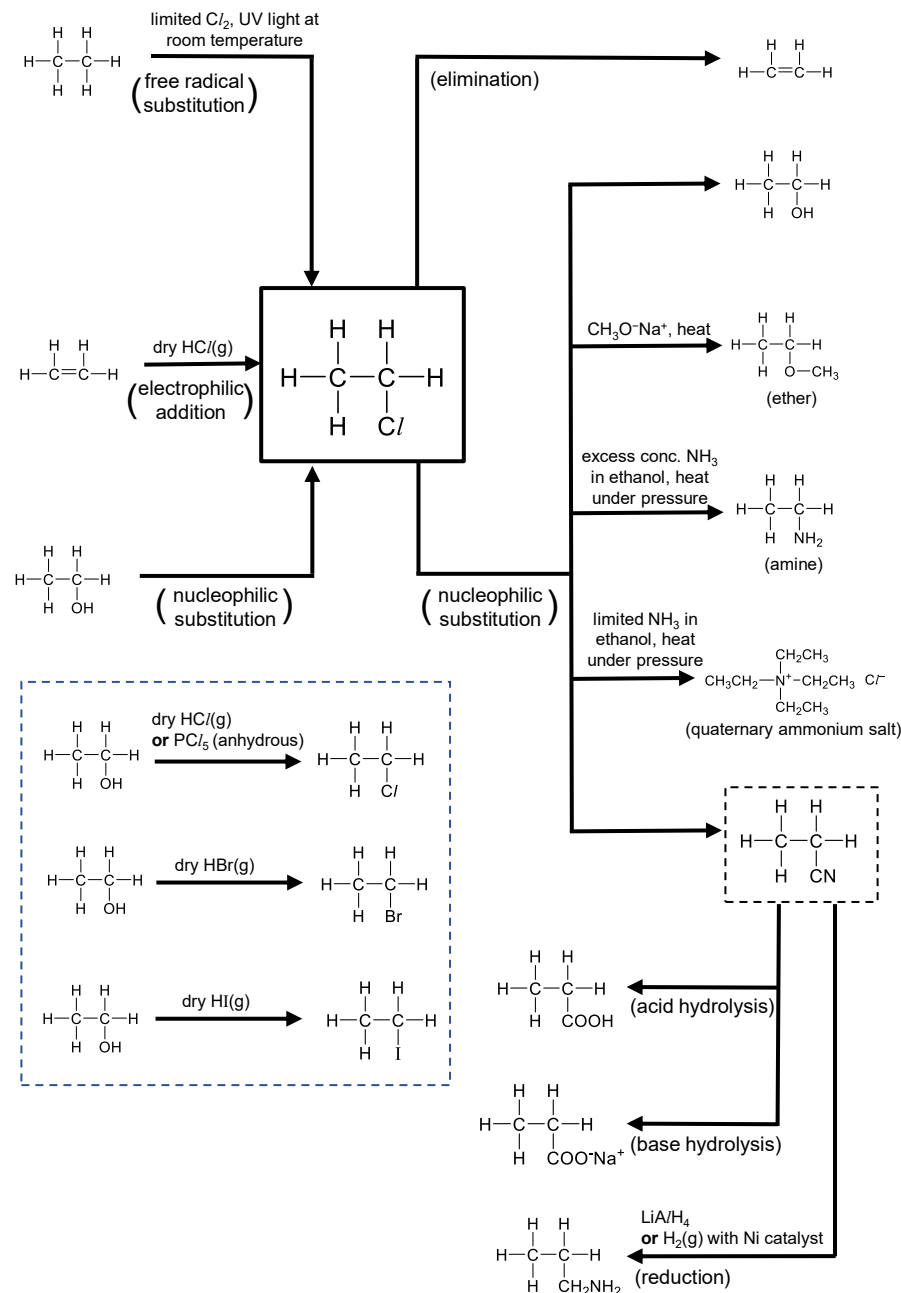
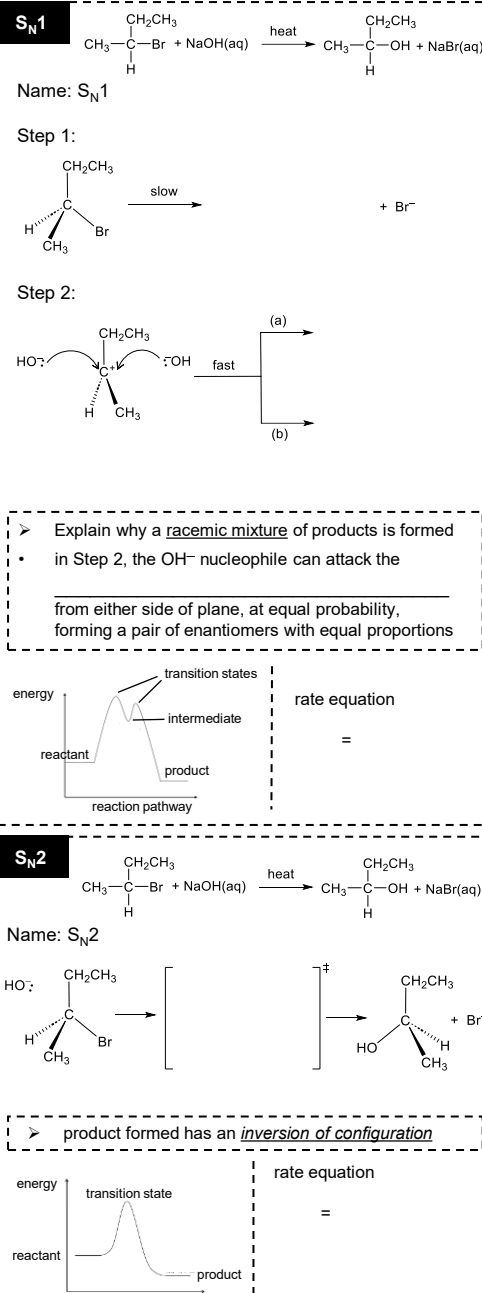


Reactions



Mechanism



S_N1 OR S_N2 ?

State whether the following halogenoalkanes undergo S_N1 / S_N2

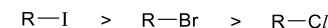
CH_3X	1°	2°	3°	allylic	benzylic	phenyl
$H-C-X$	CH_3-C-X	$CH_3-CH-CH_3$	$CH_3-C(CH_3)_2-X$	$CH_2=CH-CH_2-X$	CH_2X	X
S_N2						X

Factors determining S_N1 or S_N2

- 1 _____ about the electron-deficient C
- 2 _____ of carbocation intermediate

Relative reactivities

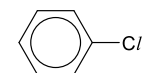
The relative rates of nucleophilic substitution (both S_N1 and S_N2) in RX are:



Using relevant data from the *Data Booklet*, explain why.

- $BE(C-Cl) = \text{_____ kJ mol}^{-1}$; $BE(C-Br) = \text{_____ kJ mol}^{-1}$; $BE(C-I) = \text{_____ kJ mol}^{-1}$
- The C-X bond strength _____ from R-Cl to R-Br to R-I.
- The _____ the C-X bond, the more _____ it breaks and hence the nucleophilic substitution takes place more rapidly.

Explain why chlorobenzene do not undergo nucleophilic substitution



1 Partial double bond character:

- p orbital of Cl _____ with the π e^- cloud of benzene ring
- lone pair of e^- on Cl can _____ into benzene ring
- the C-Cl bond is _____ (partial double bond character) and is not easily broken

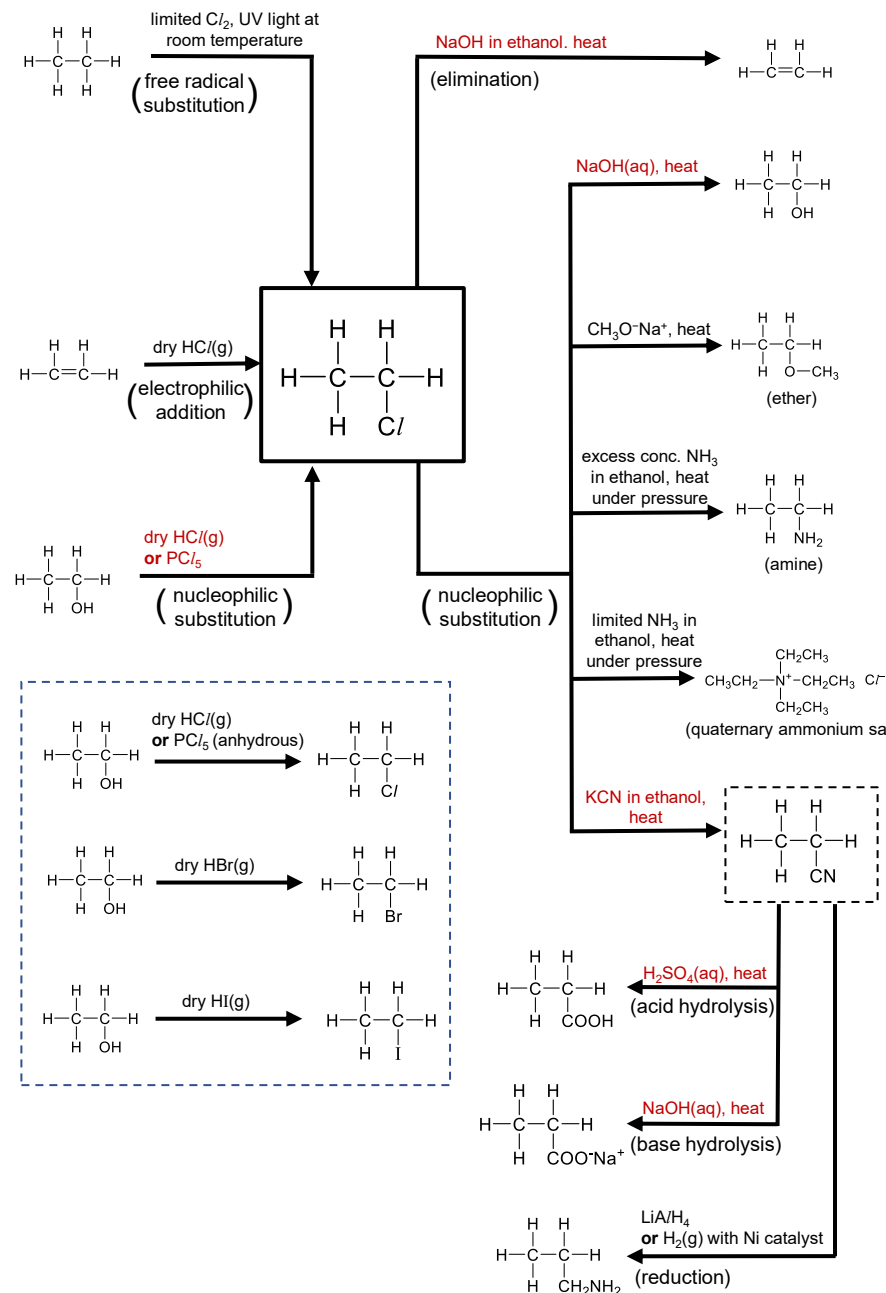
2 Steric factor:

- π e^- cloud of benzene ring repels incoming electron-rich nucleophile
- benzene ring hinders the backside attack by the nucleophile on the electron-deficient carbon

Distinguishing Tests

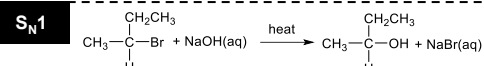
distinguish between	method/ reagents	observation
CH_3Cl , CH_3Br , CH_3I	1. add NaOH(aq) , heat 2. cool mixture 3. acidify with $\text{HNO}_3\text{ (aq)}$ 4. add $\text{AgNO}_3\text{(aq)}$	CH_3Cl : CH_3Br : CH_3I : <u>reactions which occur</u> - nu. sub of $\text{RX} \xrightarrow{\text{OH}^-}$ releases X^- - precipitation of AgX
CH_3Cl , CH_3Br , CH_3I	add ethanolic AgNO_3 and warm	CH_3Cl : CH_3Br : CH_3I : <u>reactions which occur</u> - nu. sub of $\text{RX} \xrightarrow{\text{CH}_3\text{CH}_2\text{OH}}$ releases X^- - precipitation of AgX
CH_3Br 		CH_3Br : bromobenzene : <i>Note:</i> The C-Br bond in bromobenzene is _____, has _____ character

Reactions



Mechanism

S_N1

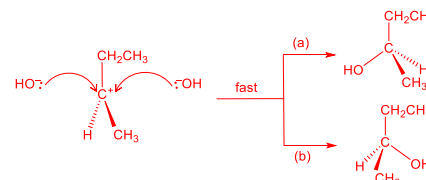


Name: S_N1

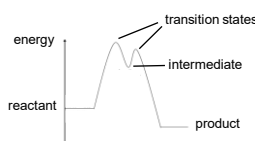
Step 1:



Step 2:



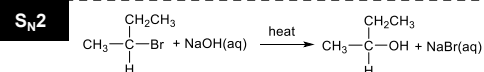
- Explain why a racemic mixture of products is formed
- in Step 2, the OH^- nucleophile can attack the sp^2 planar positively charged C from either side of plane, at equal probability, forming a pair of enantiomers with equal proportions



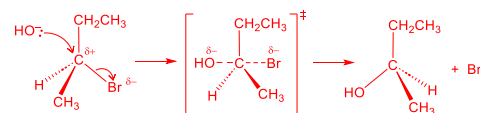
rate equation

$$\text{rate} = k[C_4H_9Br]$$

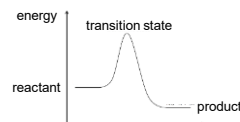
S_N2



Name: S_N2



- product formed has an inversion of configuration



rate equation

$$\text{rate} = k[C_4H_9Br][OH^-]$$

S_N1 OR S_N2 ?

State whether the following halogenoalkanes undergo S_N1 / S_N2

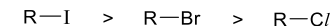
CH_3X	1°	2°	3°	allylic	benzylic	phenyl
<chem>CH3X</chem>	<chem>CH3CH2X</chem>	<chem>CH3CHXCH3</chem>	<chem>CH3C(CH3)2X</chem>	<chem>CH2=CHCH2X</chem>	<chem>c1ccccc1CX</chem>	<chem>c1ccccc1X</chem>
S_N2	S_N2	S_N1 / S_N2	S_N1	S_N1	S_N1	X

Factors determining S_N1 or S_N2

- steric hindrance about the electron-deficient C
- stability of carbocation intermediate

Relative reactivities

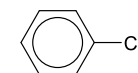
- The relative rates of nucleophilic substitution (both S_N1 and S_N2) in RX are:



Using relevant data from the *Data Booklet*, explain why.

- $BE(C-Cl) = 340 \text{ kJ mol}^{-1}$; $BE(C-Br) = 280 \text{ kJ mol}^{-1}$; $BE(C-I) = 240 \text{ kJ mol}^{-1}$
- The C-X bond strength weakens from R-Cl to R-Br to R-I.
- The weaker the C-X bond, the more easily it breaks and hence the nucleophilic substitution takes place more rapidly.

- Explain why chlorobenzene do not undergo nucleophilic substitution



1 Partial double bond character:

- p orbital of Cl overlaps with the π e^- cloud of benzene ring
- lone pair of e^- on Cl can delocalise into benzene ring
- the C-Cl bond is strengthened (partial double bond character) and is not easily broken

2 Steric factor:

- π e^- cloud of benzene ring repels incoming electron-rich nucleophile
- benzene ring hinders the backside attack by the nucleophile on the electron-deficient carbon

Distinguishing Tests

distinguish between	method/ reagents	observation
CH_3Cl , CH_3Br , CH_3I	1. add NaOH(aq) , heat 2. cool mixture 3. acidify with $\text{HNO}_3\text{ (aq)}$ 4. add $\text{AgNO}_3\text{(aq)}$	CH_3Cl : white ppt of AgCl formed CH_3Br : cream ppt of AgBr formed CH_3I : yellow ppt of AgI formed <u>reactions which occur</u> - nu. sub of $\text{RX} \xrightarrow{\text{OH}^-}$ releases X^- - precipitation of AgX
CH_3Cl , CH_3Br , CH_3I	add ethanolic AgNO_3 and warm	CH_3Cl : white ppt of AgCl formed (slowest) CH_3Br : cream ppt of AgBr formed CH_3I : yellow ppt of AgI formed (fastest) <u>reactions which occur</u> - nu. sub of $\text{RX} \xrightarrow{\text{CH}_3\text{CH}_2\text{OH}}$ releases X^- - precipitation of AgX
CH_3Br 	either of above two methods	CH_3Br : cream ppt of AgBr formed bromobenzene : no ppt formed Note: The C-Br bond in bromobenzene is <u>strengthened</u> , has <u>partial double bond</u> character